

Pleiotropy Questions:

- 1) Which of the SNPs have genome-wide significant ($p < 5 \times 10^{-8}$) associations for both traits?

- 2) Did the multivariate analyses result in additional SNPs that had genome-wide significant cross phenotype associations but that also had genome-wide suggestive ($p < 5 \times 10^{-6}$) univariate association for each trait? Which SNP(s)?

- 3) For each SNP analyzed in the mediation analysis, determine if there is a significant direct effect which is indicative of some level of biological pleiotropy. Do any of the SNPs exhibit complete mediation?

- 4) Why do some of the SNPs have negative values for the proportion mediated?

Summary table of pleiotropy results

SNP	Beta (Trait 1)	P (Trait 1)	Beta (Trait 2)	P (Trait 2)	MV ¹ P	MV ¹ Beta (Trait 1)	MV ¹ Beta (Trait 2)	ADE	ADE (P)	ACME	ACME (P)	Total Effect	Total Effect (P)	Prop Mediated	Prop Mediated (P)

¹Multivariate